(iii) If the selected first random sample is 5, then the rest of the samples are automatically selected by incrementing the value of the sampling interval (k = 10) i.e., 5, 15 , 25, 35, 45, 55, 65, 75, 85, 95.

Example:

Suppose we have to select 20 items out of 6,000. The procedure is to number all the 6,000 items from 1 to 6,000. The sampling interval is calculated as $k=\frac{N}{n}=\frac{6000}{20}=300$. Thus sampling interval = 300 denotes that for every 300 samples one sample has to be selected. The first sample is selected from the first 300 (sampling interval) samples through random selection procedures. If the selected first random sample is 50, then the rest of the samples are automatically selected by incrementing the value of the sampling interval (k=300) ie,50, 350, 650, 950, 1250, 1550, 1850, 2150, 2450, 2750, 3050, 3350, 3650, 3950, 4250, 4550, 4850, 5150, 5450, 5750. Items bearing those numbers will be selected as samples from the population.

Merits

1. This is simple and convenient method.
2. This method distributes the sample more evenly over the entire listed population.
3. The time and work is reduced much.

Demerits

1. Systematic samples are not random samples.
2. If N is not a multiple of n, then the sampling interval (k) cannot be an integer, thus sample selection becomes difficult.

8.1.2 Sampling and Non-Sampling Errors:

A sample is a part of the whole population. A sample drawn from the population depends upon chance and as such all the characteristics of the population may not be present in the sample drawn from the same population. The errors involved in the collection, processing and analysis of the data may be broadly classified into two categories namely,

(i) Sampling Errors
(ii) Non-Sampling Errors

(i) Sampling Errors

Errors, which arise in the normal course of investigation or enumeration on account of chance, are called sampling errors. Sampling errors are inherent in the method of sampling. They may arise accidentally without any bias or prejudice. Sampling Errors arise primarily due to the following reasons:

(a) Faulty selection of the sample instead of correct sample by defective sampling technique.
(b) The investigator substitutes a convenient sample if the original sample is not available while investigation.
(c) In area surveys, while dealing with border lines it depends upon the investigator whether to include them in the sample or not. This is known as Faulty demarcation of sampling units.

(ii)Non-Sampling Errors

The errors that arise due to human factors which always vary from one investigator to another in selecting, estimating or using measuring instruments(tape, scale)are called Non-Sampling errors.It may arise in the following ways:

(a) Due to negligence and carelessness of the part of either investigator or respondents.
(b) Due to lack of trained and qualified investigators.
(c) Due to framing of a wrong questionnaire.
(d) Due to apply wrong statistical measure
(e) Due to incomplete investigation and sample survey.

8.1.3 Sampling distribution

Definition 8.3

Sampling distribution of a statistic is the frequency distribution which is formed with various values of a statistic computed from different samples of the same size drawn from the same population.

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